

sort, uniq, cut, tr & wc Cheat Sheet

Small tools that each do one job to text, and get powerful when you chain them with pipes.

Using them

```
sort file      uniq file      cut -d: -f1 file    wc -l file      ← give a file, or pipe text in
tr 'a-z' 'A-Z' < file  ← tr only reads piped/redirected input, never a file name
```

sort

Option	What it does	Option	What it does
-n	Sorts as numbers (so 10 comes after 9)	-k 3	Sorts by field 3 (-k3, 3 = only field 3)
-r	Reverses the order (biggest first)	-t ', '	Fields are split at commas, not spaces
-h	Sorts sizes like 2K 5M 1G	-u	Keeps one copy of each line
-v	Version order (v1.9 before v1.10)	-o file	Writes to a file (can be the input file)

uniq, cut, tr and wc

Option	What it does	Option	What it does
uniq -c	Puts a count in front of each line	uniq -d	Shows only lines that are repeated
uniq -u	Shows only lines that appear once	uniq -i	Ignores upper/lower case
cut -d: -f1	Field 1, fields split at :	tr a-z A-Z	Lowercase to uppercase
cut -f2,5	Fields 2 and 5 (default split = tab)	tr -d '\r'	Deletes characters (Windows line ends)
cut -f3-	Field 3 to the end of the line	tr -s ' '	Squeezes runs of spaces into one
cut -c1-10	Characters 1 to 10 of each line	tr ', ' '\n'	Turns commas into new lines
wc -l	Counts lines (-w words, -c bytes)	wc -L	Length of the longest line (GNU)

cut treats every single delimiter as a split, so two spaces in a row make an empty field: squeeze with **tr -s ' '** first, or use **awk**.

Chaining them: the top-10 pattern

```
cut -d' ' -f1 access.log | sort | uniq -c | sort -rn | head
1. pull out the IP  2. group same lines  3. count each group
4. biggest count first  5. keep the top 10
```

Reading the output

```
$ ... | sort | uniq -c | sort -rn | head -3
  412 203.0.113.9 ← count, then the line it counted (412 requests from this IP)
   87 198.51.100.4
   12 192.0.2.77
$ wc /etc/passwd /etc/group
  38   61 2045 /etc/passwd ← lines, words, bytes, file name (always in that order)
  64   64  904 /etc/group
 102  125 2949 total ← a total line appears when you give more than one file
```

Handy recipes

Command	Does
sort -u names.txt	Sorted list with duplicates removed
du -sh * sort -h	Folders by size, smallest to largest
sort -t, -k3,3 -n data.csv	Sorts a CSV by its 3rd column as numbers
cut -d: -f7 /etc/passwd sort uniq -c	How many users have each login shell
sort a.txt b.txt uniq -d	Lines in both files (if neither has duplicates)
tr -s ' ' '\n' < f sort uniq -c sort -rn	Word frequency count
wc -l < file	Just the number, no file name (for scripts)
ls wc -l	How many files are in this folder

Remember: **uniq** only merges lines that are **next to each other**, so always **sort** first. Sorting numbers? Add **-n**, or 100 will sort before 9.